Fede Cotognini

Technical Sound Designer | Unreal Engine · Wwise · C++

SUMMARY

Technical Sound Designer with 3+ years in game audio, specialising in systemic, gameplay-reactive audio in Unreal Engine 5. Experienced implementing native UE5 audio: Blueprint-driven logic, MetaSounds, Sound Cues, Sound Classes, Submixes, attenuation, concurrency, and surface/material-driven systems. Background in Wwise/FMOD, C++, and C#. Comfortable taking audio features from initial design through implementation, debugging, optimisation, and polish. Remote collaborator with shipped indie titles.

EXPERIENCE

Technical Sound Designer - One More Ritual

(Unreal Engine 5 | Blueprint | C++ | Private Repository) 2025 - Present

Remote (Project near release)

- Implement gameplay-reactive audio systems in C++ using UE's native audio framework (Sound Cues, MetaSounds, Sound Classes, Submixes), without middleware.
- Built a footstep system for a first-person character without visible animations, using a state machine, timer-driven playback, and LineTrace-based surface detection.
- Debug audio issues using Visual Studio Debugger, DrawDebugSphere, and AddOnScreenDebugMessage.
- Designed reusable patterns for 40+ remaining gameplay events (Player, Weapon, Ability, Enemy, UI).

Sound Designer & Audio Programmer - Heartbeats

(Unity | FMOD | C#) 2024 - 2025

Remote (Shipped title)

- Owned the full audio pipeline and system architecture for a twin-stick shooter, developing a centralised C# API for all gameplay and UI audio events.
- Built a custom FMOD StudioListener wrapper enabling real-time 3D listener movement for fixed-camera gameplay, improving spatial audio accuracy.
- Integrated persistent volume controls using VCA buses and PlayerPrefs, ensuring audio settings reliability across user sessions.
- Collaborated with a cross-disciplinary team of 7 (designers, artists, programmers) to translate creative requirements into technical solutions.

PERSONAL PROJECT

Audio Pipeline Toolkit - Python • FFmpeg 2026 - Open Source

github.com/Cotognomo/audio-toolkit

- Built a command-line toolkit to automate repetitive audio pipeline tasks: batch conversion, asset validation, batch renaming, and CSV QC report generation.
- Published open-source on GitHub with documentation and usage examples.

SKILLS

Engines: Unreal Engine 5, Unity

Languages: C++, C#, Python (working knowledge), Blueprint

Audio Middleware: Wwise (Authoring Tool, events, RTPC, soundbanks), FMOD

Tools: Ableton Live, REAPER, Git, GitLab, Visual Studio, FFmpeg

EDUCATION

Bachelor of Arts - BA, Creative Audio Production and Sound Engineering - Catalyst - Institute for Creative Arts and Technology 2018 - 2021

High School Diploma, Drums and Percussion - Accademia di Musica Moderna di Milano 2010 - 2016

CERTIFICATION

- C++ Programming for Beginners
- Foundational C# with Microsoft
- Advanced Audio System in FMOD Studio
- AAA Game Audio with Unreal Engine 5 & Wwise
- Sound Designer for Video Games